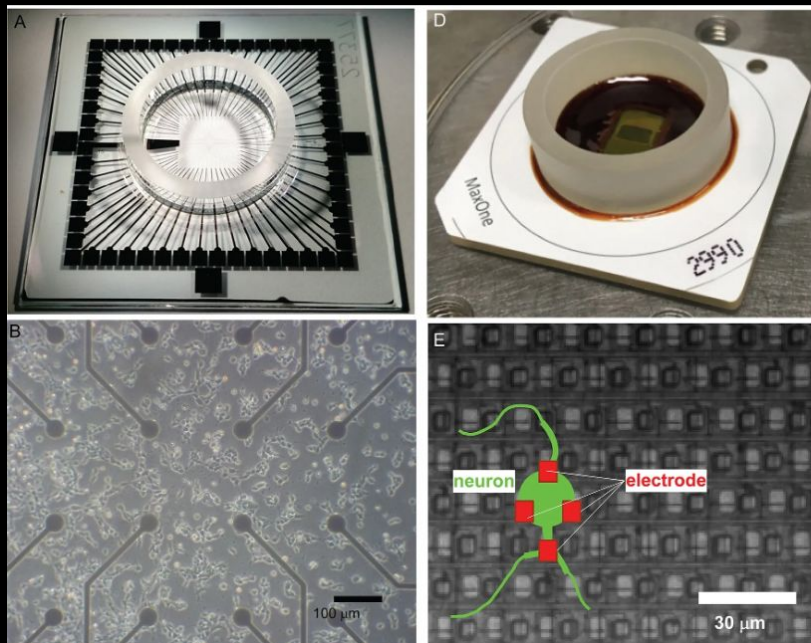


Hive Mind:

*Enabling Self-Maintenance for Small Batch
Neural Cell Cultures*

Zeon Systems Hackathon 2026

Problem



Neurons grown on Microelectrode Array (MEA)

- Media evaporates at a high rate, requiring constant replenishment and monitoring
- Cells are extremely sensitive to environmental changes
- Researchers become bottlenecked by the efforts of monitoring small batches of cells

Culture Node

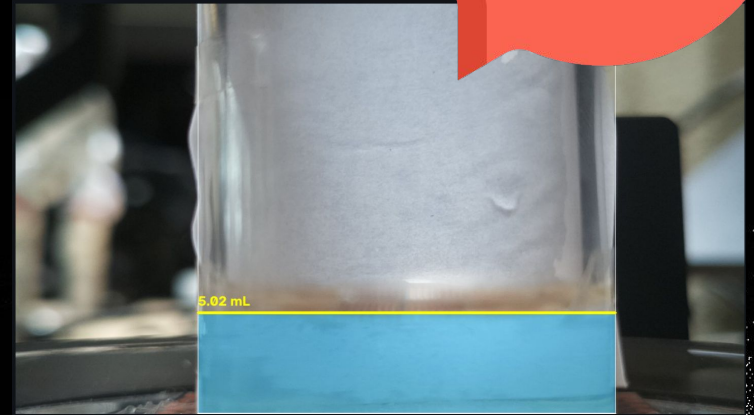
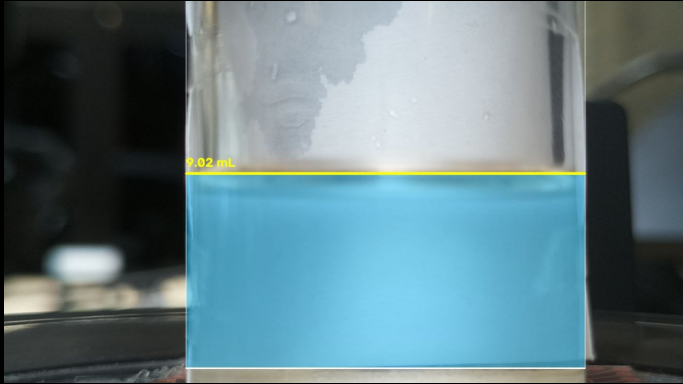
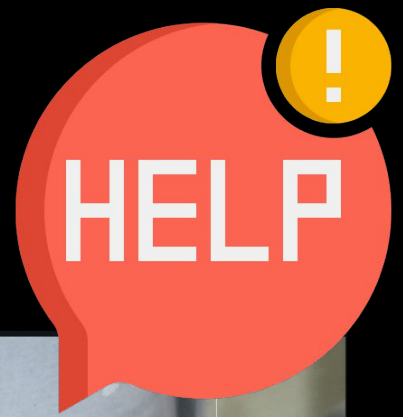


Neuron cell culture on a microelectrode array

Sensors and Illumination

Battery, Raspberry Pi

Level Monitoring



Raspberry Pi Monitoring Service

0.5.0 OAS 3.1

/openapi.json

REST interface for Raspberry Pi health checks, camera capture, liquid level, and Trinket NeoPixel control.

status

GET	/	Root	
GET	/health	Health	

camera

GET	/camera	Camera Status	
GET	/camera/image	Camera Image	

level

GET	/level	Level	
GET	/level/config	Level Config	
PUT	/level/config	Save Level Config	
POST	/level/config/reset	Reset Level Config	
GET	/level/stream	Level Stream	
GET	/level/tune	Level Tune	

led

GET	/led	Led Status	
POST	/led/on	Led On	
POST	/led/off	Led Off	
POST	/led/flash	Led Flash	

Hive



30 Culture Nodes per Hive



Hive Overwatch

Neuron Monitoring Service

0.1.0

OAS 3.1

/openapi.json

Local REST interface for neuron monitoring workflows.

status

GET

/

Root

GET

/health

Health

cultures

GET

/cultures

List Cultures

POST

/cultures/register

Register Culture

GET

/cultures/storage

Culture Storage

POST

/cultures/storage/retrieve/{id}

Retrieve Culture From Storage

POST

/cultures/storage/insert/{id}

Insert Culture Into Storage

GET

/cultures/{id}/level

Culture Level

overwatch

GET

/overwatch/levels

Overwatch Levels

POST

/overwatch/refill/{id}

Refill Culture

jobs

GET

/jobs

List Jobs

●

 Company Name

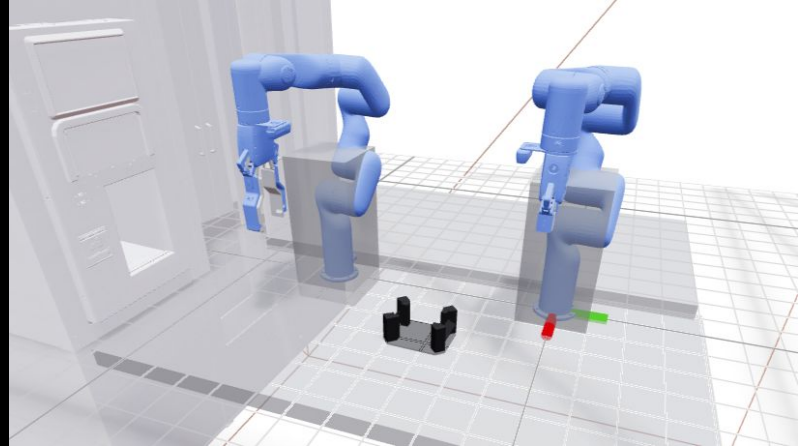
■

Year

Hive



External Workflow for Media Replenishment



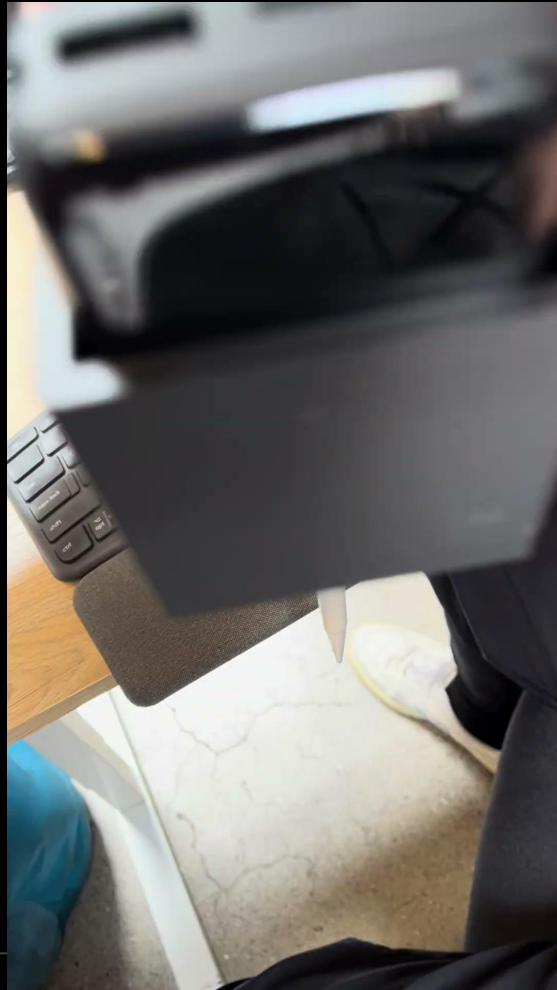
Node Loading



Node Retrieval



Level Detection



Next Time...

Expand the
sensor suite

Integrate more
workflows

Shrink the culture
node

Improve power
management